

# Amnesia by Design: On Temporal Traceability in Transformer Representations

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**Abstract.** Words do not preserve fixed meanings across time. `gay@1920` and `gay@1980` are different semantic objects, yet transformers treat them as the same token in the same geometric space. Masked language modeling rewards correct token prediction rather than geometric consistency across time, so temporal traceability is not directly optimized.

We introduce a minimal benchmark with known ground-truth semantic trajectories to demonstrate this limitation under controlled conditions, and formalize Timeformer, a family of token-time conditioning designs of increasing complexity. Temporal conditioning substantially improves neighborhood drift, yet no architecture fully matches the planted trajectories across diagnostics. The failure of our most complex extension further localizes the problem: when a word develops coexisting senses, mean pooling collapses both into a centroid that represents neither.

The results bear directly on any application that requires word meaning at a specific period — from historical semantic analysis to knowledge-intensive tasks over time-sensitive corpora — and suggest that progress on temporal traceability requires training objectives aligned with trajectory geometry.

**Keywords:** Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

## 1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; today it describes a social media post reaching millions of users. Whether the shift of *gay* from *cheerful* to its contemporary sense involved movement toward or away from stigmatized terms [5] is an empirical question that requires `gay@1920` and `gay@1980` as directly comparable objects in a shared geometric space.

Contextualized embeddings disambiguate senses — but via local sentential context, not period. A model correctly represents *virus* differently in a sentence with *malware* and *code* versus one with *infection* and *bacteria*, using the neighboring tokens as evidence. The period is irrelevant for this disambiguation. The limitation appears when one wants to query `token@time` without a supporting sentence: `gay@1920` as a directly inspectable object, not as an embedding that

only separates from `gay@1980` when forced by local context. For diachronic analysis — in which direction did the meaning move, how close was it to stigmatized terms — we need the directly inspectable object.

Masked language modeling masks a token and asks the model to reconstruct it from its neighbors. To do this well, the model learns local context; it has no direct incentive to place the same token in period-appropriate regions of the representation space. Temporal traceability — `virus@1900` and `virus@2020` occupying different neighborhoods — would require a training signal aligned with geometric consistency across time. The MLM objective does not provide such a signal directly.

Demonstrating this limitation requires a controlled environment where the correct answer is known. Natural-language corpora entangle semantic change with frequency, genre, and subcorpus shifts, making it hard to attribute a representational effect to architecture rather than data. The minimal benchmark we introduce isolates this effect before moving to open-domain corpora.

Against this benchmark we formalize Timeformer, a family of token-time conditioning designs of increasing complexity. The comparison isolates the contribution of each architectural decision: whether any temporal signal helps, whether joint token-time conditioning outperforms global period injection, and whether causal prototype memory adds anything beyond conditioning. The controlled environment makes each question answerable without corpus confounders.

The paper makes four contributions: it establishes temporal traceability as a geometric property distinct from period-label recoverability; shows in a minimal controlled benchmark that standard transformer training supports this property only indirectly; formalizes Timeformer as a diagnostic and architectural framework; and identifies mean-prototype aggregation as a second bottleneck.

## 2 Related Work

Lexical semantic change detection (LSCD) is the dominant computational framing for meaning shift. Diachronic word embeddings [5] learn one distributional space per time slice, align independently trained models, and measure movement in neighborhood structure. We share the intuition that semantic change is geometric, but not the architecture: LSCD yields a scalar comparison between spaces rather than a jointly parameterized `token@time` representation. Standard benchmarks formalize this task [8, 9], while also inheriting natural-corpus confounds such as frequency and genre drift; shared tasks for Spanish show that manual annotation is required even for modern corpora [11]. Our controlled benchmark plants trajectories so that traceability can be attributed to model design rather than corpus composition.

Temporal neural language models avoid separate spaces by conditioning one model on time. Prior work uses timestamps for knowledge-intensive QA, masked language modeling, or period-stratified downstream evaluation [3, 7, 6]. These results show that temporal conditioning can improve period-appropriate predic-

tion, but they do not test whether the same token occupies different neighborhoods at different periods in a shared space.

Linear probes provide another partial view. A period probe measures whether time is recoverable from a hidden state, not whether a token has moved to a different semantic neighborhood. We therefore report probe accuracy as auxiliary evidence and treat neighborhood movement as the primary criterion for traceability.

Finally, semantic change may involve bifurcation rather than monotonic drift. Giulianelli et al. show that words can develop coexisting senses [4]; mean-pooled representations then collapse both senses into a centroid that represents neither. Our stable, drifting, and bifurcating trajectories test this distinction directly, and the memory extension exposes where mean-pooled prototypes fail.

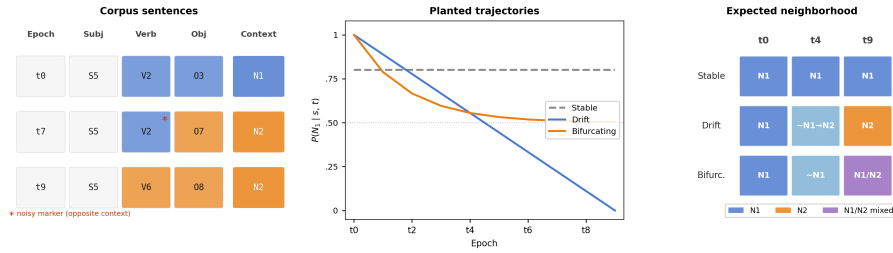
### 3 Controlled Temporal Benchmark

The benchmark gives each subject a known temporal trajectory, making it possible to ask directly whether a model’s representation of a subject at  $t_0$  has moved toward the expected neighborhood by  $t_9$ . This distinguishes it from natural-language LSCD benchmarks, where ground truth is often annotated or inferred post-hoc and corpus confounds are hard to separate from genuine semantic change.

The corpus uses a subject–verb–object (SVO) grammar: every sentence consists of one subject, one verb, and one object. Two semantic neighborhoods,  $N_1$  and  $N_2$ , partition the vocabulary of 8 verbs and 8 objects:  $N_1$  uses  $\{V1, \dots, V4\}$  and  $\{O1, \dots, O4\}$ ;  $N_2$  uses  $\{V5, \dots, V8\}$  and  $\{O5, \dots, O8\}$ . A sentence generated from  $N_1$  therefore looks like **S5 V2 O3**, while one from  $N_2$  looks like **S5 V6 O8**; the subject token is the same in both cases, but its co-occurring words differ. Intuitively, the intended meaning of subject  $s$  at period  $t$  is how often it appears in  $N_1$  sentences; formally,  $P(N_1 | s, t)$  is the probability that a sentence is drawn from  $N_1$ . The vocabulary contains 30 subjects; each subject’s trajectory is determined by how  $P(N_1 | s, t)$  evolves across the 10 periods  $t_0$  through  $t_9$ .

Figure 1 (center) shows the three planted trajectory classes. Stable subjects maintain an approximately constant  $P(N_1 | s, t)$  throughout. Drift subjects follow a monotonic trajectory from near-exclusive  $N_1$  ( $P(N_1 | s, t) \approx 1$  at  $t_0$ ) to near-exclusive  $N_2$  ( $P(N_1 | s, t) \approx 0$  at  $t_9$ ): a drifting subject that appears in sentences like **S5 V2 O3** early on will appear in sentences like **S5 V6 O8** by the final period. Bifurcating subjects begin in  $N_1$  and move toward a mixed state ( $P(N_1 | s, t) \approx 0.5$ ) where both neighborhoods appear with equal frequency, approximating two coexisting senses. The benchmark contains 10 subjects per class.

Each local marker is drawn from the opposite neighborhood 25% of the time (Figure 1, left), so the model cannot recover the planted neighborhood from co-occurring words alone — it must use subject identity together with the period. The `true_context` label is withheld from training and used only for evaluation.



**Fig. 1.** Benchmark design overview. *Left:* SVO sentences for a drifting subject across three periods; an  $N_1$  verb appearing in an  $N_2$  sentence at  $t7$  (marked \*) illustrates the deliberate noise in local markers. *Center:* planted  $P(N_1 | s, t)$  trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across  $t0$ – $t9$ . *Right:* expected nearest-neighbor behavior for a traceable representation.

The synthetic benchmark is deliberate: it isolates architectural effects that natural corpora confound with frequency, genre, and composition shifts. By planting trajectories with known  $P(N_1 | s, t)$ , we can ask directly whether a model’s geometry matches the generating process; the scope of these mechanistic diagnostics is revisited in the Discussion.

The benchmark supports several evaluation conditions. The standard split uses the same 75% marker accuracy as training. Two harder splits force the verb, or both verb and object, to come from the opposite context. An ambiguous split sets fidelity to 50%, so local markers carry no reliable signal; only subject identity and period remain. Two additional conditions test temporal behaviour directly and are described in Section 5: the contrastive set presents the same sentence under two different period labels to check whether changing only the period changes the preferred context; the continuation set uses periods  $t8$ – $t9$ , held out from MLM training (which covers  $t0$ – $t7$ ), to test generalisation.

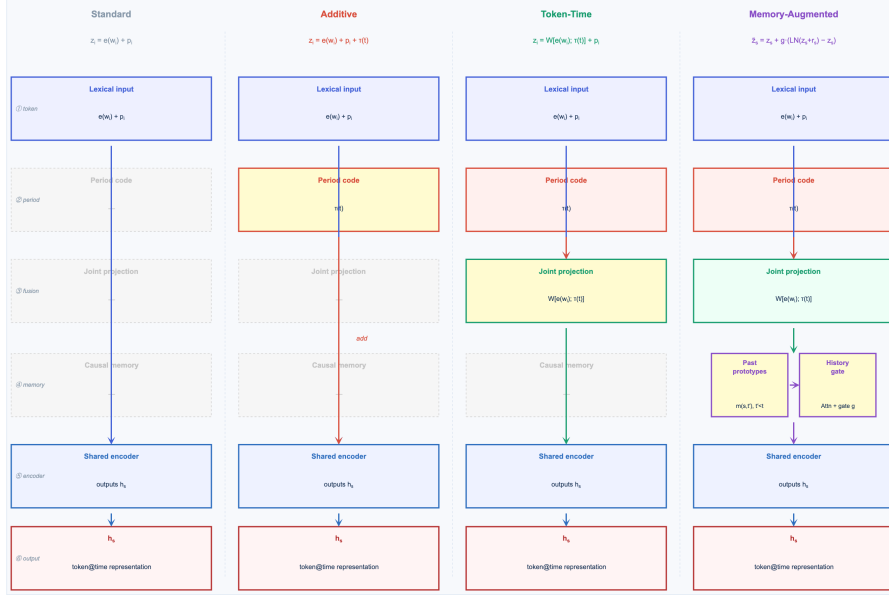
## 4 Timeformer Architecture

Timeformer is designed to make a subject representation depend explicitly on the queried period. We study this through a controlled comparison: Standard has no period input, Additive injects a global period vector, Token-Time exposes token identity and period jointly before self-attention, and the Memory-Augmented extension adds causal historical prototypes before the encoder.

All models follow the same outer pipeline based on the Transformer encoder [10]. A sentence is tokenized as  $[\text{CLS}] \text{ S V O } [\text{SEP}]$ . Each token  $w_i$  is mapped to an input embedding  $z_i \in \mathbb{R}^d$ . The sequence  $(z_1, \dots, z_n)$  is processed by a Transformer encoder, and the final hidden state at the subject position,

$$(h_1, \dots, h_n) = \text{Encoder}(z_1, \dots, z_n), \quad h_s = h_{\text{subj}},$$

is the `token@time` representation inspected by the diagnostics. All models are trained with masked language modeling (MLM, [2]): some tokens are masked,



**Fig. 2.** Controlled design comparison. **Standard:** Transformer baseline with no temporal signal. **Additive:** temporal baseline that adds the same period vector to every token. **Token-Time:** Token-Time Transformer using joint token-time projection. **Memory-Augmented:** extension of Token-Time with Prototype Memory and Temporal Attention. New components are highlighted in yellow; dashed grey boxes mark inactive components.

and the model reconstructs them from the remaining tokens and, when available, the period input. The variants differ only in how they build  $z_s$  before the encoder.

The temporally informed models use a shared period vector  $\tau(t) \in \mathbb{R}^d$ , a learned projection of sinusoidal features of  $t/T$ , where  $T$  is the number of training periods. It is distinct from the position encoding  $p_i$ , which marks subject, verb, or object position. Additive and Token-Time differ in how they combine  $\tau(t)$  with token identity; Memory-Augmented then adds causal memory. Figure 2 summarizes the comparison.

#### 4.1 Baseline: Standard Transformer

Standard is the null hypothesis: no period signal is available, so any temporal behavior must arise from changes in the observed words alone.

$$z_i = e(w_i) + p_i.$$

## 4.2 Baseline: Additive Time-Conditioned Transformer

The Additive baseline asks whether knowing the current period is enough. It adds the same period vector  $\tau(t)$  to every token:

$$z_i = e(w_i) + p_i + \tau(t).$$

Now S5 V2 O3 at  $t0$  and  $t9$  are distinguishable: the subject receives  $\tau(0)$  or  $\tau(9)$ . The same temporal change is also applied to the verb and object, so Additive tells the whole sentence which period it belongs to but does not explicitly bind period to each token’s identity.

## 4.3 Token-Time Transformer

The Token-Time Transformer in Figure 2 asks whether exposing token identity and period jointly before self-attention is enough. Instead of adding the same period vector to all tokens, Token-Time concatenates each token embedding with  $\tau(t)$  and projects the pair:

$$z_i = W[e(w_i); \tau(t)] + p_i,$$

where  $e(w_i), \tau(t) \in \mathbb{R}^d$  and  $W \in \mathbb{R}^{d \times 2d}$ . Although  $W$  is linear, it exposes token-period inputs before self-attention and later nonlinear layers. Thus S5 at  $t0$  and S5 at  $t9$  enter the encoder as distinct period-conditioned inputs.

## 4.4 Memory-Augmented Extension

The Memory-Augmented extension asks whether causal history helps beyond token-time conditioning. It starts exactly like Token-Time: for a current sentence containing S5 at  $t7$ , the model first builds token-time input embeddings for [CLS] S5 V O [SEP]. It then modifies only the subject embedding before the Transformer encoder sees the sentence.

The additional information comes from Prototype Memory. The extension stores one prototype vector per subject and past period. A prototype  $m(s, t)$  is the average representation of subject  $s$  across training sentences from period  $t$ . The memory is causal: at  $t7$ , the model may consult  $\{m(\text{S5}, 0), \dots, m(\text{S5}, 6)\}$ , but not prototypes from  $t7$  or later.

Temporal attention turns this list of past prototypes into a history summary for the current subject. Let  $z_s$  be the current Token-Time subject embedding and let  $M_s$  be the matrix of available causal prototypes, one row per past period. The attention operation uses  $z_s$  as the query and  $M_s$  as keys and values:

$$r_s = \text{Attn}(z_s, M_s, M_s).$$

The vector  $r_s$  summarizes the parts of the subject’s history that are most relevant to the current period. A learned scalar gate  $g$  then decides how much this history summary should change the current subject embedding:

$$\tilde{z}_s = z_s + g \cdot (\text{LN}(z_s + r_s) - z_s).$$

Here, LN denotes layer normalization, and  $g$  is a learned scalar gate initialized at zero so that the extension reduces to Token-Time at initialization. Finally,  $\tilde{z}_s$  replaces  $z_s$  in the input sequence; verb and object embeddings are unchanged, so the subject is the only token that enters the encoder informed by causal prototype history.

## 5 Evaluation Protocol and Diagnostics

All models in the comparison chain are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is withheld from training and used exclusively for post-hoc evaluation: it tests whether the learned subject representation reflects the planted temporal trajectory. The primary evaluation target is  $h_s$ , the final hidden state at the subject position, because subjects are the lexical items whose meanings are designed to drift, remain stable, or bifurcate over time.

We use four diagnostics, each targeting a distinct aspect of temporal representation.

**D1 – Linear probe [1].** Asks whether  $h_s$  encodes the planted context label at all. The encoder is frozen; a linear classifier is trained on  $h_s$  representations extracted from the held-out test sentences of the standard (75%-fidelity) split and evaluated on the ambiguous and continuation splits. High accuracy indicates that period-appropriate context is linearly separable in the representation, but does not reveal whether that separation shifts systematically over time.

**D2 – Context drift score.** Asks whether the neighborhood of a subject shifts in the expected direction across periods. For each period and subject class, we retrieve the  $k = 10$  nearest neighbors of  $h_s$  by cosine similarity (excluding the query itself) and record the proportion assigned to  $N_1$ . A representation is traceable at period  $t$  if this proportion tracks  $P(N_1 \mid s, t)$  — the same probability that governs corpus generation at that period. For drifting subjects, the proportion should therefore decrease monotonically from  $t_0$  to  $t_9$ ; for stable subjects it should remain near its initial value; for bifurcating subjects it should decrease less steeply, as the representation splits between neighborhoods rather than migrating from one to the other.

**D3 – Contrastive diagnostic.** Asks whether the period label alone drives representational change. We present the same sentence with two different period labels and compute the sign-flip rate: the proportion of pairs for which swapping the period reverses the nearest-neighbor assignment, from  $N_1$ -dominant to  $N_2$ -dominant or vice versa. A model that ignores the period input produces a sign-flip rate of zero; a model that responds to it produces a rate above zero.

**D4 – Continuation diagnostic.** Asks whether the learned temporal mapping generalizes beyond training. We evaluate on held-out periods  $t_8$ – $t_9$ , excluded from MLM training, and operationalize the diagnostic as linear-probe accuracy on representations from these periods. A model that has internalized

the direction of change, rather than memorizing period-specific patterns, should preserve context recoverability in unseen periods.

The four diagnostics are interpreted through a fixed comparison chain. Additive vs. Standard isolates the effect of adding any temporal signal. Token-Time vs. Additive tests whether exposing token identity and period jointly before self-attention is more effective than applying the period globally as a uniform shift. Memory-Augmented vs. Token-Time tests the contribution of causal prototype memory, the mechanism most expected to affect D3 and D4.

For the Memory-Augmented extension, we include an *oracle-memory* diagnostic: learned prototypes are replaced by fixed vectors encoding the true historical  $P(N_1 | s, t)$  trajectory, with the rest of the architecture unchanged. This isolates whether the failure of learned prototypes is due to the memory pathway itself or to the aggregation mechanism; improvement under oracle prototypes indicates that the pathway is functional but the aggregation is the bottleneck.

## 6 Results

We report means and 95% confidence intervals over 31 training seeds. The main finding is that any temporal conditioning produces substantially larger neighborhood drift than no conditioning; Additive and Token-Time are not robustly distinguishable on drift score. The Memory-Augmented extension does not provide a robust improvement over the Token-Time Transformer.

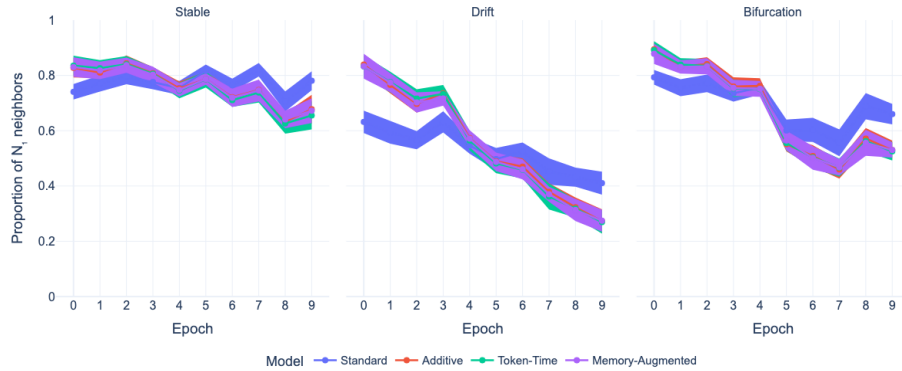
*Implementation details.* All models share the same Transformer encoder:  $d = 64$ , 4 attention heads, 2 layers, feed-forward width 128, dropout 0.1. The period encoding projects 32 sinusoidal features of  $t/T$  to  $d$  via a two-layer MLP. Training uses AdamW ( $\text{lr} = 10^{-3}$ , weight decay  $= 10^{-2}$ ) with a cosine annealing schedule, 30 epochs, batch size 64. The corpus contains 5,000 SVO sentences; MLM training uses 3,159 sentences from periods  $t0$ – $t7$ ; 763 sentences from the same periods form the held-out evaluation set; 794 sentences from  $t8$ – $t9$  are reserved for the continuation diagnostic.

### 6.1 Temporal Neighborhood Movement

Figure 3 and Table 1 show the context drift score across periods. Standard is a sanity check: its weak movement ( $\Delta = -0.221$ , 95% CI  $[-0.240, -0.203]$ ) confirms that the benchmark alone does not produce the stronger trajectories seen when period information is available. All three conditioned models reach  $\Delta \approx -0.56$ , with overlapping confidence intervals; the gap between any conditioned model and Standard is the primary geometric evidence for the effect of temporal conditioning.

The class-specific curves serve as controls (Table 1). Stable subjects show smaller movement in absolute magnitude ( $\Delta$  between  $-0.150$  and  $-0.180$  for conditioned models, versus  $\Delta \approx -0.56$  for drifting subjects), but the direction is consistently negative. This reflects a global displacement of the embedding space





**Fig. 3.** Context drift score across periods. All conditioned models move drifting subjects from  $N_1$  toward  $N_2$  substantially more than Standard; Additive, Token-Time, and Memory-Augmented produce similar trajectories. Stable and bifurcating subjects provide class-specific controls.

induced by the period vector: even subjects with flat planted trajectories are pulled along the learned temporal direction, at substantially reduced magnitude. Bifurcating subjects show intermediate drift: conditioned models reach  $t_9$  values near 0.53, consistent with the planted endpoint  $P(N_1 | s, t_9) \approx 0.5$  for the mixed state, confirming that the benchmark’s bifurcating trajectory is partially recovered but not fully resolved. The geometric diagnostic distinguishes all three trajectory classes without relying on probe accuracy alone.

## 6.2 Probe and Contrastive Diagnostics

The linear probe results reveal a distinction that drift score does not. On the standard (75%-fidelity) split, where local markers carry reliable context signal, all conditioned models outperform Standard ( $0.875 [\pm 0.005]$ ): Additive reaches  $0.892 [\pm 0.003]$ , Token-Time  $0.881 [\pm 0.005]$ , and Mem-Aug.  $0.881 [\pm 0.006]$ ; on this split Additive is above Token-Time. The ordering reverses on the ambiguous split, where local verb and object markers are uninformative: Standard reaches  $0.565 [0.561, 0.568]$ , Additive  $0.579 [0.575, 0.584]$ , Token-Time  $0.595 [0.589, 0.601]$ , and Memory-Augmented  $0.598 [0.590, 0.606]$ . The confidence intervals for Additive and Token-Time do not overlap, showing a reproducible advantage for joint token-time conditioning when only subject identity and period remain as evidence. On the continuation split, Additive and Token-Time reach the same accuracy (0.763; Table 2), consistent with the indistinguishability on drift score. Probe accuracy alone is insufficient to establish temporal traceability — it measures recoverability of the planted label, not neighborhood geometry — but the split-dependent pattern suggests that joint conditioning encodes temporal information more precisely in the subject representation, with the advantage visible only when local markers are removed.

**Table 1.** Context drift score at the first and last periods. Values are means over 31 seeds;  $\Delta = t_9 - t_0$ ; 95% CI half-widths in brackets.

| Class   | Model      | t0    | t9    | $\Delta$ [CI $_{\pm}$ ] |
|---------|------------|-------|-------|-------------------------|
| Stable  | Standard   | 0.741 | 0.781 | +0.040 [ $\pm 0.017$ ]  |
|         | Additive   | 0.827 | 0.678 | -0.150 [ $\pm 0.027$ ]  |
|         | Token-Time | 0.835 | 0.655 | -0.180 [ $\pm 0.024$ ]  |
|         | Mem-Aug.   | 0.830 | 0.672 | -0.158 [ $\pm 0.021$ ]  |
| Drift   | Standard   | 0.632 | 0.410 | -0.221 [ $\pm 0.018$ ]  |
|         | Additive   | 0.839 | 0.272 | -0.567 [ $\pm 0.019$ ]  |
|         | Token-Time | 0.833 | 0.269 | -0.563 [ $\pm 0.020$ ]  |
|         | Mem-Aug.   | 0.833 | 0.274 | -0.559 [ $\pm 0.022$ ]  |
| Bifurc. | Standard   | 0.793 | 0.660 | -0.133 [ $\pm 0.018$ ]  |
|         | Additive   | 0.895 | 0.530 | -0.365 [ $\pm 0.017$ ]  |
|         | Token-Time | 0.891 | 0.525 | -0.366 [ $\pm 0.016$ ]  |
|         | Mem-Aug.   | 0.878 | 0.530 | -0.348 [ $\pm 0.015$ ]  |

**Table 2.** Main traceability diagnostics (means, 31 seeds). Drift: context drift score  $\Delta$  for drifting subjects; Flip: contrastive sign-flip rate (D3); Ambig./Cont.: linear-probe accuracy on  $h_s$ . 95% CI in brackets.

| Model      | Drift                  | Flip                  | Ambig.                | Cont.                 |
|------------|------------------------|-----------------------|-----------------------|-----------------------|
| Standard   | -0.221 [ $\pm 0.018$ ] | 0.000                 | 0.565 [ $\pm 0.004$ ] | 0.730 [ $\pm 0.004$ ] |
| Additive   | -0.567 [ $\pm 0.019$ ] | 0.602 [ $\pm 0.053$ ] | 0.579 [ $\pm 0.005$ ] | 0.763 [ $\pm 0.003$ ] |
| Token-Time | -0.563 [ $\pm 0.020$ ] | 0.624 [ $\pm 0.048$ ] | 0.595 [ $\pm 0.006$ ] | 0.763 [ $\pm 0.004$ ] |
| Mem-Aug.   | -0.559 [ $\pm 0.022$ ] | 0.587 [ $\pm 0.051$ ] | 0.598 [ $\pm 0.008$ ] | 0.755 [ $\pm 0.005$ ] |

The contrastive sign-flip rates are similar across conditioned models: Standard 0.000, Additive 0.602 [0.549, 0.655], Token-Time 0.624 [0.576, 0.673], Mem-Aug. 0.587 [0.536, 0.638]. Confidence intervals overlap, confirming that all conditioned models respond to period information at roughly the same rate in contrastive pairs.

### 6.3 Memory Diagnostics

The Memory-Augmented extension does not produce a robust gain over the Token-Time Transformer. Across 31 seeds, learned prototypes reach a continuation accuracy of 0.755 [0.750, 0.760], marginally below Token-Time at 0.763 [0.759, 0.766] ( $\Delta = -0.008$ ). The oracle diagnostic shows the limits of this architecture: replacing learned prototypes with fixed vectors encoding the true  $P(N_1 | s, t)$  trajectory yields a continuation of 0.769 [0.761, 0.777] and a sign-flip rate of 0.694 [0.576, 0.812], versus 0.624 for Token-Time. The sign-flip improvement ( $\Delta = +0.070$ ) is larger than the continuation improvement ( $\Delta = +0.007$ ), but both CIs are wide. The dissociation has a direct explanation: the MLM objective optimizes token prediction, not trajectory geometry; perfect historical prototypes can therefore improve contrastive behavior — which depends on lo-

**Table 3.** Memory diagnostics (means, 31 seeds). Continuation (D4): linear-probe accuracy on held-out periods  $t8-t9$ ; Flip: contrastive sign-flip rate (D3); 95% CI half-widths in brackets.

| Model             | Cont.               | Flip                | $\Delta$ cont. vs. Token-Time |
|-------------------|---------------------|---------------------|-------------------------------|
| Token-Time (ref.) | 0.763 $[\pm 0.004]$ | 0.624 $[\pm 0.048]$ | –                             |
| Mem-Aug. learned  | 0.755 $[\pm 0.005]$ | 0.587 $[\pm 0.051]$ | –0.008 $[\pm 0.006]$          |
| Mem-Aug. oracle   | 0.769 $[\pm 0.004]$ | 0.694 $[\pm 0.059]$ | +0.007 $[\pm 0.007]$          |

cal period sensitivity — but cannot substitute for a training signal that directly rewards geometric continuity across unseen periods.

## 7 Discussion and Conclusion

No architecture fully matches the planted trajectories across diagnostics. Conditioned models improve neighborhood drift substantially — Standard reaches  $\Delta = -0.221$  while all conditioned models cluster near  $\Delta \approx -0.563$  — but even the oracle diagnostic, which provides a perfect memory signal, lifts continuation accuracy only to 0.769. This ceiling points less to a single architectural failure than to the MLM objective, which rewards token prediction rather than geometric consistency across periods. Traceability emerges as a side effect, and the results show how far this incidental signal goes.

The form of conditioning matters less than its presence. Additive and Token-Time are statistically indistinguishable on drift score across 31 seeds; Token-Time shows a reproducible advantage only on the ambiguous probe (0.595 vs. 0.579, non-overlapping CIs), where local markers are uninformative and only subject identity and period remain. This suggests that joint token-time conditioning encodes temporal information more precisely in the subject representation, even when the aggregate neighborhood trajectory is similar. The convergence of two architecturally distinct designs toward the same drift performance points to the training objective as the binding constraint, not the conditioning mechanism.

The Memory-Augmented extension points to an additional bottleneck. For bifurcating subjects, which develop two coexisting senses in later periods, a mean prototype lies between clusters and represents neither well. The oracle diagnostic confirms that the memory pathway is useful in principle: replacing learned prototypes with fixed vectors encoding the true  $P(N_1 | s, t)$  trajectory improves the sign-flip rate by +0.070 over Token-Time (0.694 vs. 0.624). The results are therefore consistent with an aggregation bottleneck: mean-pooled representations cannot preserve the internal structure of a word’s sense distribution.

These conclusions are mechanistic diagnostics, not claims about open-domain diachronic language. The probe-level advantage of Token-Time over Additive is statistically stable but modest in magnitude (1.6 percentage points) and does not appear on the drift score; it may not generalize beyond the controlled setting of fixed subject positions and small vocabulary. The diagnostic suite was

co-designed with the architecture family, limiting its role as an independent evaluation for models outside this design space.

Two directions follow directly. Memory mechanisms that preserve the internal structure of historical representations — rather than collapsing them to a single prototype per subject and period — are the most direct next step for the Memory-Augmented extension. At the level of the main comparison, the convergence of Additive and Token-Time on drift performance is worth investigating: attention heads may learn to suppress the temporal signal during self-attention, neutralizing differences in input conditioning. Analysis of attention patterns stratified by period would make this testable, and could inform whether progress on traceability requires changes to the objective, the architecture, or both.

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